Remember

$F = C - P + 2$ Gibbs phase rule

Osmotic pressure $RT \ln g_1 x_1 = - \underline{V}_1 \Delta P$

Surface tension $P^{IN} - P^{OUT} = \frac{2\sigma}{r}$

Droplets $\underline{V}^L \left(\frac{2\sigma}{r} + \cancel{P^V - P^{SAT}} \right) = RT \ln \frac{P^V}{P^{SAT}}$

Bubbles $\frac{2\sigma}{r} = P^{SAT} - P^{ATM}$

Multicomponent Vapour Liquid Equilibrium (VLE)

If we can deal with binary mixtures we can also do calculations for mixtures with more components, by combining the binary parameters

We will therefore focus on binary mixtures

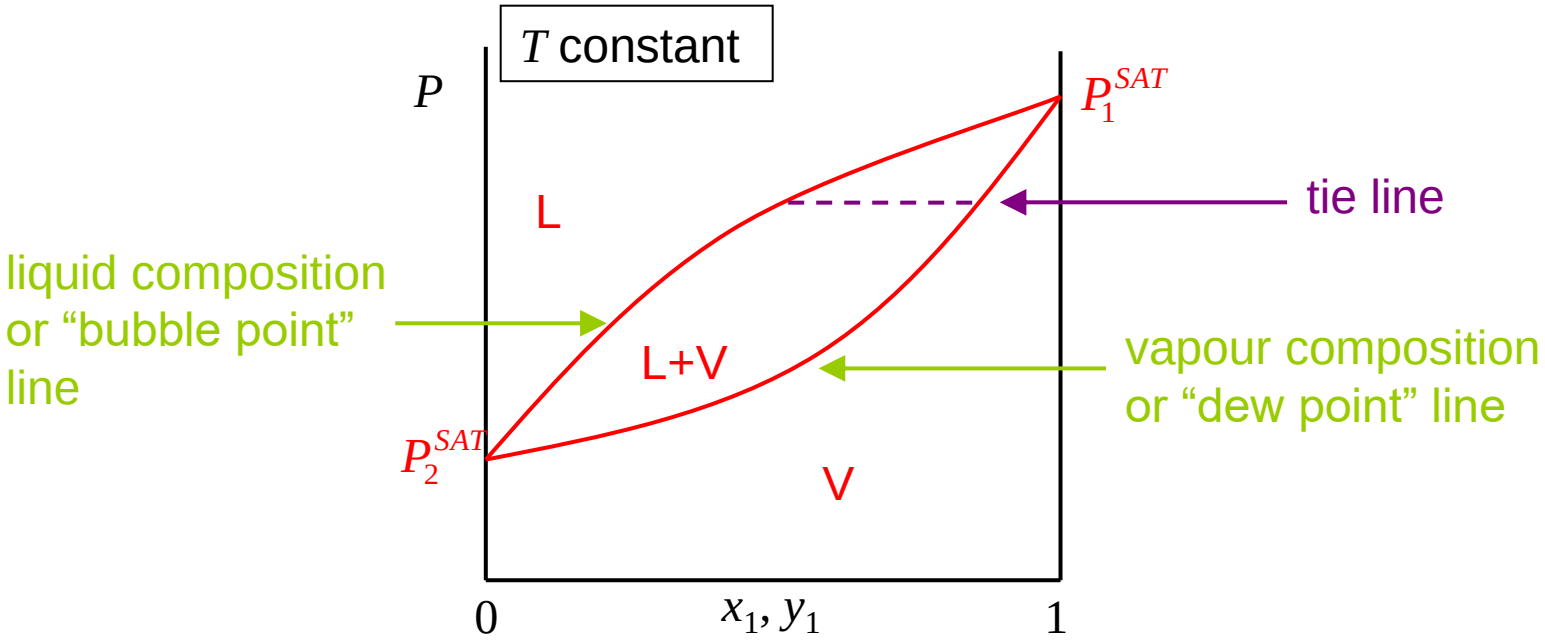
Binary Mixtures at VLE

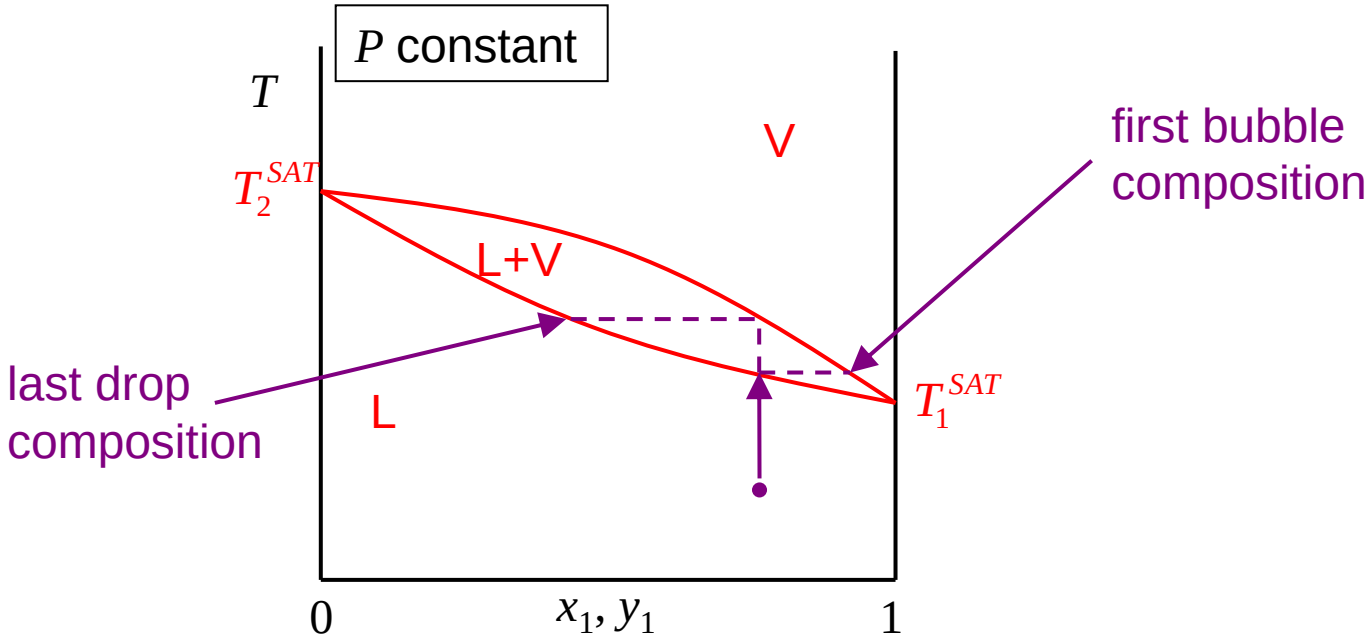
The phase rule gives; (n.b. assuming a flat interface)

$$F = C - P + 2 = 2 - 2 + 2 = 2$$

For example, if we fix the temperature we are free to vary the pressure **or** the composition (but not both independently)

This behaviour is represented graphically on “reduced phase diagrams”


T constant
P
P_1^{SAT}
L
tie line
liquid composition
or “bubble point”
line
L+V
vapour composition
or “dew point” line
P_2^{SAT}
V
0
x_1, y_1
1


P constant
T
V
first bubble
composition
T_2^{SAT}
L+V
last drop
composition
L
T_1^{SAT}
0
x_1, y_1
1

The criteria for vapour-liquid equilibrium are:

$P^l = P^v$ (assuming a flat interface)

$T^l = T^v$

$\mu_i^l = \mu_i^v$ (or equivalently $\overline{f}_i^{\,l} = \overline{f}_i^{\,v}$)

because $\mu_i^l = G_i^{SAT} + RT \ln \dfrac{\overline{f}_i^{\,l}}{P_i^{SAT}}$, etc.

now have more than one component

These equations, together with the conservation of species:

$$Lx_i + Vy_i = x_i^0 \qquad i = 1,2,.....,C$$

where L = mole fraction of liquid

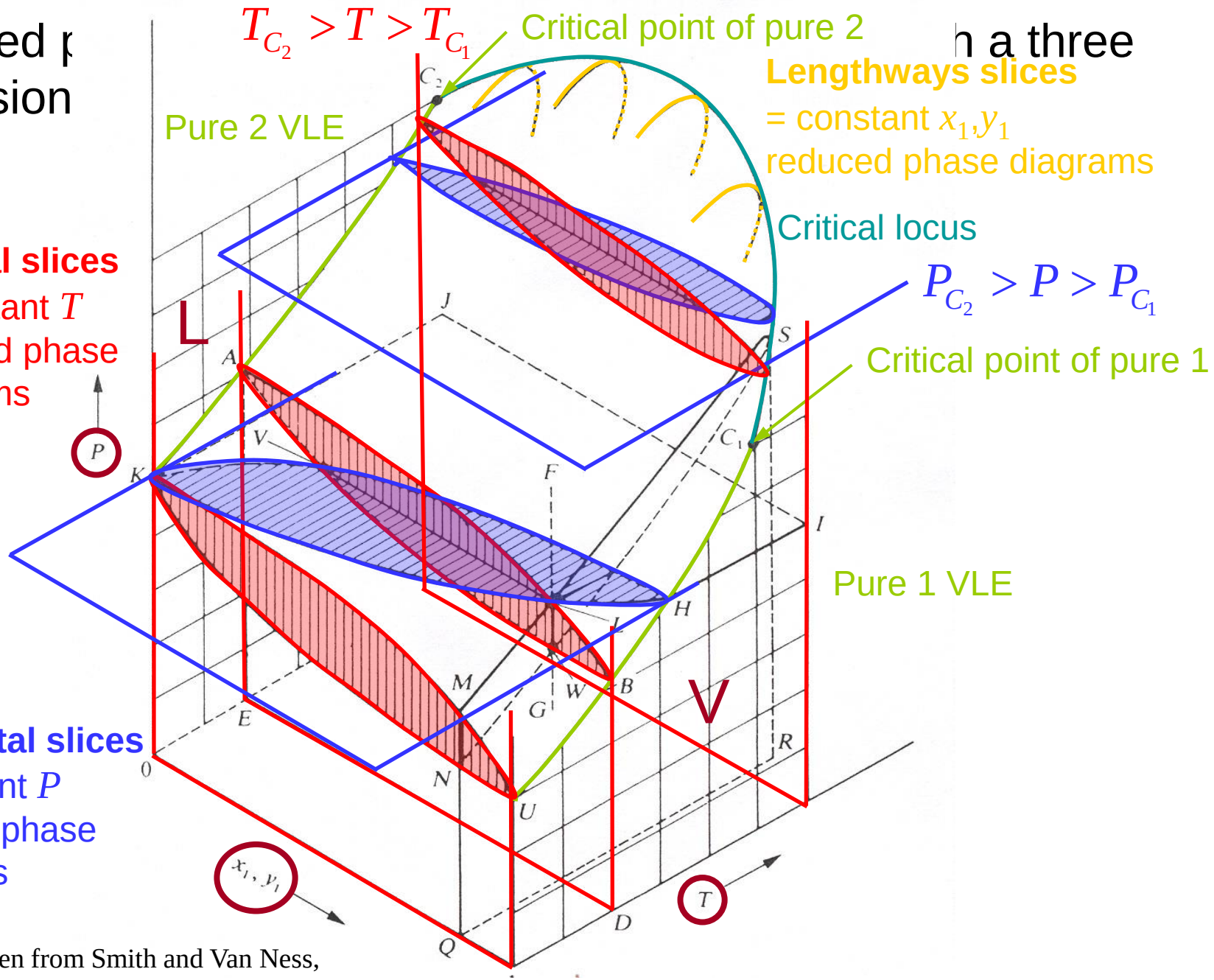
V = mole fraction of vapour

x_i^0 = mole fraction of i which would be present if all the vapour were condensed to liquid

and $\sum x_i = 1; \quad \sum y_i = 1; \quad L + V = 1$

provide the entire theoretical framework for all VLE problems

Reduced p … n a three dimension



Vertical slices
= constant T
reduced phase diagrams

Horizontal slices
= constant P
reduced phase diagrams

[Diagram taken from Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics", MacGraw Hill.]

3D plot

VLE Using Activity Coefficient Models

To solve $\mu_i^l = \mu_i^v$ we need chemical potentials in terms of x_i and y_i

μ_i^v may be derived from an EOS

For the liquid we may use either an Activity Coefficient Model or an EOS consistent with that for the vapour.

We will assume that the Lewis-Randall rule is valid for the vapour phase, i.e the vapour phase behaves as an ideal mixture, but not necessarily an ideal gas.

(Note: this is not a necessary assumption - we could use a vapour mixture EOS, but this would be unpleasant)

$\mu_i^l = \mu_i^v$ then becomes:

mixture

pure vapour

$$\mu_i^v = \underline{G}_i^0 + RT \ln \frac{\overline{f}_i^{\,v}}{P^0} = \underline{G}_i^0 + RT \ln \frac{y_i f_i^{\,v}}{P^0}$$

pure IG at T, P^0

$$\mu_i^l = \underline{G}_i(T,P) + RT \ln \gamma_i x_i = \underline{G}_i^0 + RT \ln \frac{f_i^{\,l}}{P^0} + RT \ln \gamma_i x_i$$

$$\Rightarrow \quad \boxed{\gamma_i x_i f_i^{\,l} = \overline{f}_i^{\,v} = y_i f_i^{\,v}}$$

Furthermore, we can relate the (pure) liquid phase fugacity, f_i^l to the value at saturation conditions using the Poynting correction:

$$f_i^l \approx f_i^{SAT}(T)\exp\left\{\frac{\left(P-P^{SAT}\right)\underline{V}^l}{RT}\right\}$$

At same T, P as mixture

Thus we have:

$$\gamma_i x_i \underbrace{\phi_i^{SAT} P^{SAT}}_{f_i^{SAT}(T)} \exp\left\{\frac{\left(P-P^{SAT}\right)\underline{V}^l}{RT}\right\} = y_i \underbrace{\phi_i^v P}_{f_i^v(T,P)}$$

$\Leftarrow$ EOS

At low pressures, approximate that $\phi_i^{SAT} = \phi_i^v = 1$, and ignore the Poynting correction, giving:

$$\boxed{\gamma_i x_i P_i^{SAT} = y_i P}$$

Most common starting point for estimates of VLE

Raoult's Law (Ideal Mixtures)

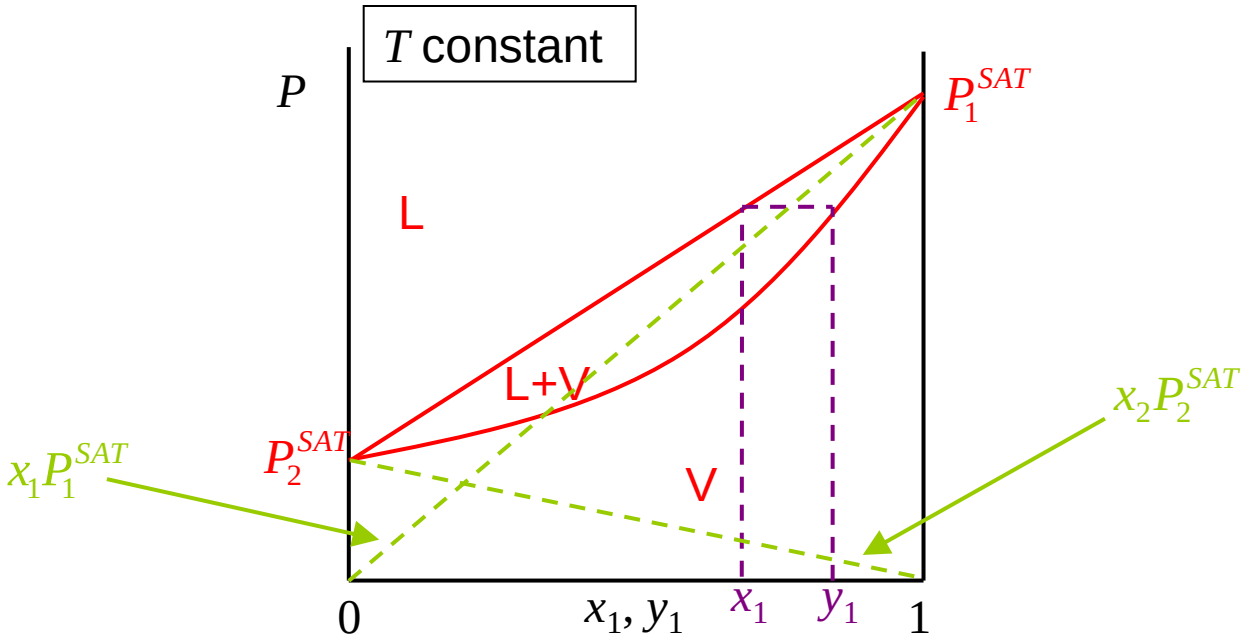
is obtained by applying the following additional simplification:

$\gamma_i = 1$ i.e. ideal (liquid) mixture

Now $x_i P_i^{SAT}(T) = y_i P$

and $\boxed{\sum x_i P_i^{SAT}(T) = \sum y_i P = P}$

For a binary mixture:



Raoult's law ≡ IM at low P

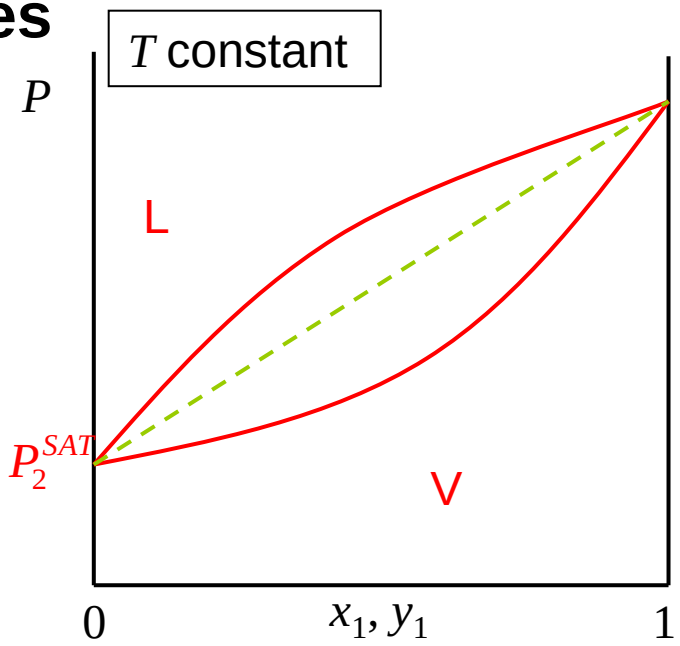
Real Mixtures

Few mixtures show simple Raoult's law behaviour

Positive deviations from Raoult's law may occur (i.e. $P > \sum x_i P_i^{SAT}$) if the activity coefficients, γ_i, are greater than 1

The converse is true for negative deviations (i.e. $P < \sum x_i P_i^{SAT}$)

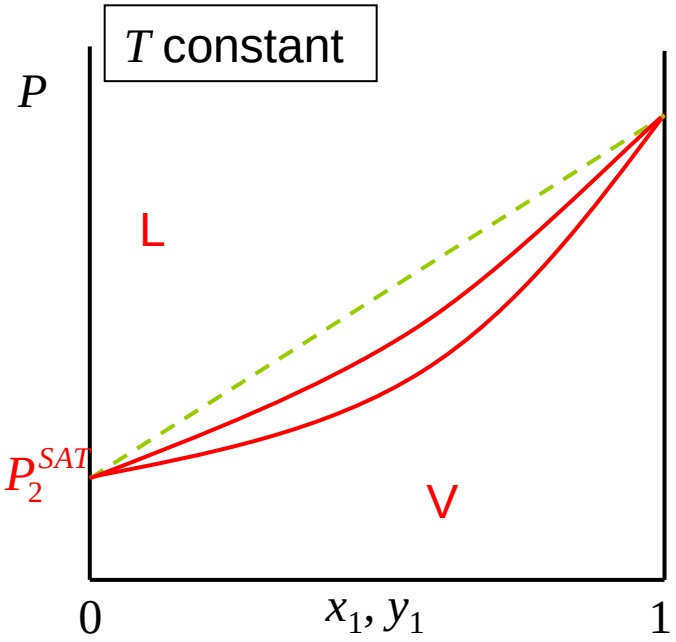
Binary mixtures



Raoult's law

+ve deviation
liquid line is above
Raoult's law line

$\gamma_i s > 1$
material wants to get
out of liquid phase



Raoult's law

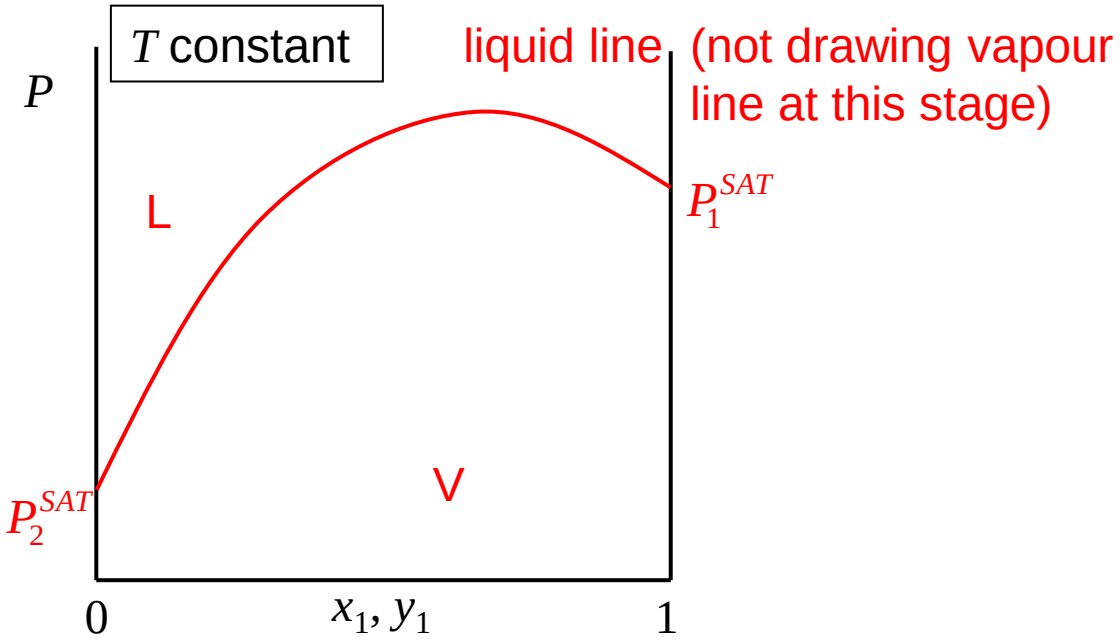
-ve deviation
liquid line is below
Raoult's law line

$\gamma_i s < 1$
material is attracted to
liquid phase

Azeotropes (we will consider only binary mixtures)

Azeotropes are turning points in the P-x curves

$$\frac{dP}{dx_1} = 0 \quad \Rightarrow \quad \text{max or min in } P \text{ (not at } x_1 = 1 \text{ or } 0)$$



(i.e. $f_i^v = P$)

Neglecting the Poynting correction and assuming $\phi_i = 1$ in the vapour phase {this will be OK at low *P*}, we have for VLE:

$$\gamma_i x_i P_i^{SAT} = y_i P$$ common starting point for VLE

and $$P = x_1\gamma_1 P_1^{SAT} + x_2\gamma_2 P_2^{SAT}$$ since binary

$$\therefore \quad \frac{dP}{dx_1} = \gamma_1 P_1^{SAT}(1 + x_1 \frac{d\ln\gamma_1}{dx_1}) + \gamma_2 P_2^{SAT}(-1 + x_2 \frac{d\ln\gamma_2}{dx_1})$$

Using the Gibbs-Duhem equation, we obtain:

$$x_1 \frac{d\ln\gamma_1}{dx_1} + x_2 \frac{d\ln\gamma_2}{dx_1} = 0$$

$$\frac{dP}{dx_1} = (\gamma_1 P_1^{SAT} - \gamma_2 P_2^{SAT})(1 + x_1 \frac{d\ln\gamma_1}{dx_1}) = 0$$ at azeotrope

$$\frac{dP}{dx_1} = (\gamma_1 P_1^{SAT} - \gamma_2 P_2^{SAT})(1 + x_1 \frac{d \ln \gamma_1}{dx_1}) = 0$$

Thus for a minimum or maximum in *P*, we have either

$$\gamma_1 P_1^{SAT} = \gamma_2 P_2^{SAT}$$ azeotrope

or $$\frac{d \ln \gamma_1}{dx_1} = -\frac{1}{x_1}$$ $\Rightarrow$ liquid-liquid separation (see later lecture)

Recall the VLE expression:

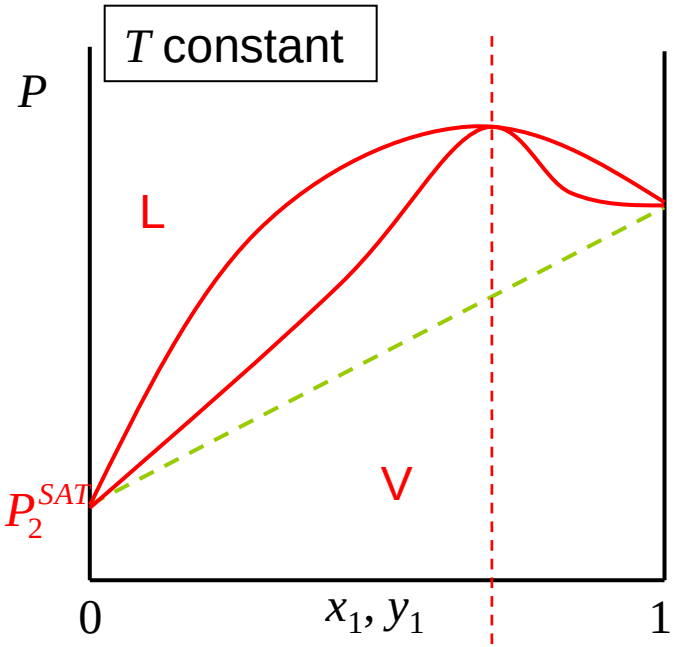
$$y_1 P = y_1 (x_1 \gamma_1 P_1^{SAT} + x_2 \gamma_2 P_2^{SAT}) = x_1 \gamma_1 P_1^{SAT}$$

Substituting $\gamma_1 P_1^{SAT} = \gamma_2 P_2^{SAT}$

$\Rightarrow \quad y_1 = x_1$ i.e., the vapour and liquid have the same composition for a binary azeotrope

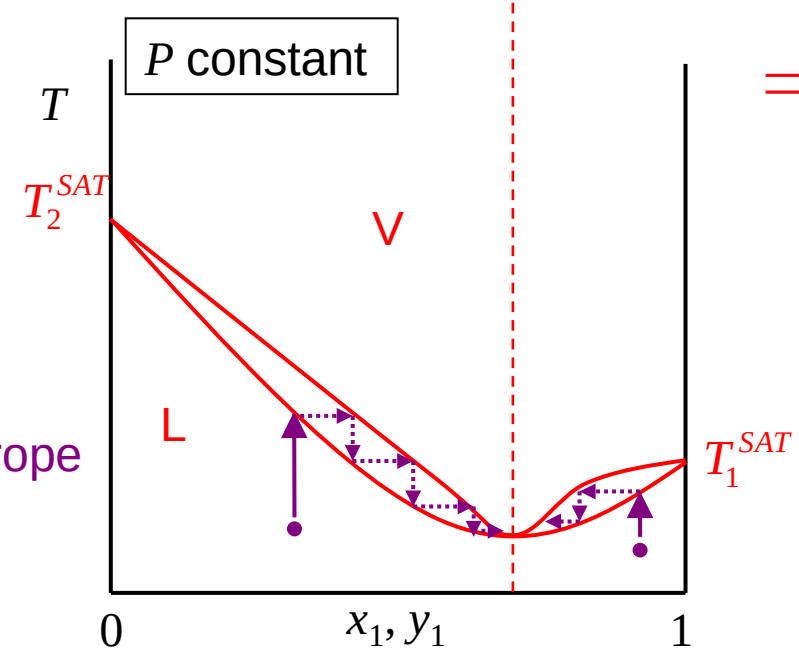
n.b. this is an extra constraint for the Gibbs phase rule

$\Rightarrow$ only one degree of freedom at the azeotrope



Raoult's law

+ve deviation
from Raoult's law
$\Rightarrow$ max boiling P



$\Rightarrow$ min boiling T

distill to the azeotrope

n.b. to the right of the azeotrope, the vapour phase has higher mole fraction of the less volatile component than the liquid phase

Note:

1) Maximum boiling temperature (minimum boiling pressure) azeotropes are also possible.

2) Azeotropes are annoying for distillation. To break an azeotrope, we need to add an extra degree of freedom. How?

extra component, e.g. benzene/ethanol/water